

Class 07 — Homework

Bong Study Hub — Foundation Batch 2026 · Class 07: Operators & Expressions

Class 01's LOGIC, Made Literal

Estimated time: 30–45 minutes **Total:** 30 marks · **Optional Challenge:** up to 3 bonus marks **Tools needed:** pen and paper, or a simple document. A computer with Python is helpful for checking your work, but not required to answer any question. **Prerequisite:** Classes 05, 06, and 07. No `if`, `input()`, loops, or compound assignment (`+=`) are needed anywhere in this assignment.

What This Assignment Is About

In class, we built the toolkit that turns Class 01's abstract `LOGIC` box into real code: **arithmetic operators** (`+`, `-`, `*`, `/`, `//`, `%`) that compute, **comparison operators** (`==`, `!=`, `<`, `>`, `<=`, `>=`) that ask true/false questions and produce booleans, and **logical operators** (`and`, `or`, `not`) that combine those booleans. We also met the single most common beginner trap: confusing `=` (store) with `==` (ask).

This assignment asks you to **trace code carefully, not just recall facts about it**. Several questions give you an expression or a short program and ask exactly what it produces — order of operations and the `=`/`==` distinction both reward careful, patient reading.

Instructions

- **Answer in your own words** where a question asks for an explanation.
- For "predict the result" or "trace the code" questions, write the **exact** value(s) Python would produce, including `True`/`False` capitalized exactly as Python writes them.
- **No `if`, `input()`, loops, or `+=` appear anywhere in this homework** — if your answer to any question involves one of those, simplify it back down to operators and expressions.
- You may refer to your **Student Notes** while answering.
- If you're unsure, **explain your reasoning anyway** rather than leaving a blank — partial thinking is worth more than an empty space.

Section A — Core Concepts

Question 1 — What Is an Operator? (2 marks)

In your own words, **explain what an operator is**, and what the word **"operand"** means.

Question 2 — / vs // (2 marks)

Explain the difference between `/` and `//` in Python, using one short example of each.

Question 3 — What Does % Give You? (2 marks)

In your own words, explain what the `%` operator produces, with one example.

Question 4 — Comparisons and Data Types (2 marks)

What data type does every comparison operator (`==` `!=` `<` `>` `<=` `>=`) always produce? Explain how this connects to Class 06.

Question 5 — = vs == (2 marks)

Explain the difference between `=` and `==` in Python. Why is mixing them up such a common — and risky — mistake?

Section B — Reading and Reasoning About Code

Question 6 — Predict the Output (3 marks)

```
a = 12
b = 5
print(a + b)
print(a - b)
print(a * b)
print(a / b)
```

Write the exact output, line by line, in order.

Question 7 — Order of Operations (3 marks)

```
print(3 + 4 * 2)
print((3 + 4) * 2)
```

Write the exact output of both lines, and explain in one sentence why they differ.

Question 8 — Predict the Comparisons (3 marks)

```
x = 10
print(x == 10)
print(x != 5)
print(x < 5)
print(x >= 10)
```

Write the exact output, line by line, in order.

Question 9 — Predict the Logical Results (3 marks)

```
p = True
q = False
print(p and q)
print(p or q)
print(not q)
```

Write the exact output, line by line, in order.

Question 10 — Find the Bug (3 marks)

A student wanted to check whether `age` equals `18` and store the true/false answer in `result`. Here is what they actually wrote:

```
age = 18
result = age = 18
print(result)
```

What does this actually print? (Hint: it will not crash — that's what makes this bug dangerous.)

What did the student probably mean to write instead, and what would that corrected version print?

Section C — Class Connection & Your Own Program

Question 11 — Connecting Class 01 and Class 07 (2 marks)

Fill in the blank below, using ideas from today's class:

```
LOGIC (Class 01)
↓
_____ (Class 07)
```

Briefly explain, in one sentence, **how today's operators make Class 01's `LOGIC` box literal.**

Question 12 — Write Your Own Expressions (3 marks)

Imagine you're planning a small trip. **Write a short program** (on paper is fine) with:

- At least **two number variables** (e.g. a budget and a cost).
- **One arithmetic expression** combining them (e.g. money left over).
- **One comparison** that produces a boolean (e.g. whether you're within budget).
- **One logical expression** combining two comparisons (e.g. whether the trip is both affordable and long enough).

Print each result.

Optional Challenge — Bonus (up to 3 marks)

This section is entirely optional and will not affect your base grade.

Trace the program below very carefully — it combines arithmetic, order of operations, comparison, and logical operators. **Write the exact output**, in order.

```
a = 17
b = 5
c = 2

result1 = a // b + c * 2
result2 = a % b == 2
result3 = result2 and (result1 > 5)

print(result1)
print(result2)
print(result3)
```

Submission Format

Choose whichever is easiest for you:

- **Option A:** Handwrite your answers, then photograph or scan them.
- **Option B:** Type your answers in a simple document.
- **Option C:** Submit as a simple PDF.

Please label each answer with its question number.

Before You Submit — Self-Check

- ☐ I answered all 12 questions.
- ☐ For every "predict the output" or "trace the code" question, I wrote the exact value(s), including ``True`/`False`` capitalized correctly.
- ☐ Where unsure, I explained my reasoning instead of leaving a blank.
- ☐ I did not use ``if``, ``input()``, loops, or ``+=`` anywhere.
- ☐ I reviewed my work before submitting.